NOLI · DATA ENGINEER BUSINESS CASE · SECTION 1

A trusted attribution architecture

End-to-end design from paid and owned channels to a single, governed number — using a multi-tool stack, not a single vendor.

Audience: Senior Data Engineer · Principal Engineer   15 minutes
Stack: Fivetran · Databricks / Delta · Unity Catalog · Snowflake · dbt · Airflow · Power BI

Sidhartha Thogari · Data Engineer interview

HOW WE WILL USE 15 MINUTES

The brief, answered in order

01 **Diagnose**
Why Meta, Google Ads and Email cannot agree — grain, identity, windows, UTMs, vendor claim.

02 **Architecture**
ELT path from sources to Power BI. Medallion + a restricted hash / PII store.

03 **Stack trade-offs**
Fivetran, Databricks, Delta / Unity Catalog, Snowflake, dbt, Airflow, Power BI — why and downside.

04 **Model & logic**
Canonical attribution model. Logic in the warehouse, never in the BI tool.

05 **Trust at scale**
Contracts, tests, freshness, reconciliation. Incremental design for TBs/day.

PROBLEM STATEMENT

Marketing does not have a number. It has three stories.

Inconsistent attribution across Meta, Google Ads and Email. UTMs on URLs are missing or malformed.

Vendor-claimed conversions

Meta and Google both take credit for the same order. Email tools report 'revenue' on last click to a newsletter.

Broken campaign contract

utm_source = fb / Facebook / meta / ig / paid_social. Same campaign, four keys. Sometimes no UTM at all.

Unstable grain

One report is sessions. One is users. One is orders. One is clicks. None are labelled.

Logic in the last mile

Analysts fix mismatches in Power BI or Excel. Tomorrow the patch is a different number.

Five reasons the numbers diverge — before we pick tools

Identity	Cookie, email, customer_id, device. Logged-out quiz vs logged-in checkout. No stitch → double count or drop.
Attribution window & model	1-day click vs 7-day click vs 28-day view. Last-click vs linear vs platform data-driven. Must be explicit columns, not opinions.
Event vs commercial grain	Clicks and sessions are not orders. Refunds, cancellations and partial returns change 'attributed revenue' after the fact.
UTM + landing hygiene	Missing, mixed case, aliases, leftover internal params. Paid landing pages without a tagging standard.
Two sources of truth	Platform UI (Meta Ads Manager) vs warehouse. If we hide the vendor number, trust dies. If we mix it into the KPI, trust dies.

DESIGN PRINCIPLES

How I would improve trust — before drawing boxes

ELT, not ETL soup

Land immutable raw payloads. Transform in the warehouse with tested SQL. Hash PII on the way in so raw email is not a free-for-all.

One grain, many models

Order-level attribution is the commercial grain. Touchpoints stay at event grain. Models are columns or a model_name key — not competing dashboards.

Logic lives once

dbt (and a thin Python layer for hashing / identity). Not Fivetran pre-build transforms. Not Power BI measures that redefine net revenue.

Contract the tags

Campaign naming + UTM dictionary + validation at the edge. Warehouse maps aliases; marketing owns the standard.

Reconcile, don't hide

Vendor-claimed vs canonical. Freshness, volume, null-UTM rate. Tickets become dashboards.

Privacy by construction

Hashed keys in Gold. Restricted mapping store. Consent flags. RTBF. Clean-room-ready shares for partners.

Sources → ingest → lakehouse / warehouse → governed Gold → BI

Azure. Multi-tool. Airflow is the conductor — not a second transformation engine.

SOURCES

Paid
Meta Ads
Google Ads

Owned
Email / CRM
(Klaviyo etc.)

Site & commerce
GA4 / pixel
Orders, refunds
Consent / CMP

INGEST

Fivetran
Managed SaaS CDC
Meta, Google, email,
commerce, GA4 export

Events / files
Auto Loader / APIs
for pixel + exceptions

Airflow
Sync → hash → dbt
→ tests → refresh

PROCESS & MODEL

Bronze · Delta / Snowflake RAW
Immutable JSON/tables · schema evolution
Audit cols · Fivetran metadata

Silver · dbt + Databricks Python
UTM map · identity stitch
Hash PII · SCD2 campaigns

Gold · dbt marts
fct_touchpoint · fct_order
fct_attribution · dim_*

SERVE

Snowflake
Serving + shares
Clean rooms

Unity Catalog + Snowflake RBAC
Lineage, masks
Row filters

PII vault
Hash map
DE-only

CONSUME

Power BI
Certified
semantic model

dbt docs
Metrics &
exposures

AI / ML
Feature tables
Genie / models

Across the top: Entra ID · Key Vault (hash pepper) · Unity Catalog + Snowflake policies · Airflow SLAs · cost/freshness alerts. Reverse ETL later — after identity is clean.

Technology · Why · Downside

I would not rip-and-replace Noli's modern data stack. I would stop tools from overlapping.

Technology	Why it earns a place	Downside I would manage
Fivetran	Managed, reliable connectors for Meta, Google Ads, email, GA4, commerce. Fast time-to-Bronze. Ops-light incremental sync.	Cost at volume. Vendor dependency. Weak as a transform layer — keep it extract + load.
Databricks	Python/Spark for UTM parsing, identity, PII hashing, TB-scale semi-structured, ML feature tables for AI enablement.	Needs engineering and tuning. Job-cluster cost if left on. Do not re-implement dbt models in notebooks.
Delta + Unity Catalog	ACID, time travel, schema evolution, lineage, RBAC, column masks. Auditability when attribution is challenged.	Platform complexity. Governance only works if we actually register tables and stop ad-hoc personal schemas.
Snowflake	JD-aligned serving warehouse. BI-friendly, sharing and clean rooms with brands / L'Oréal-class partners.	Another compute bill if we also run Databricks. Decide a system of record for Gold — I pick Snowflake for finance & marketing KPIs.
dbt	SQL models, tests, docs, contracts, exposures. The only place 'attributed_revenue' is defined.	Overlap if Fivetran transformations or Power BI DAX duplicate the same logic. Discipline required.
Airflow	Cross-platform orchestration: Fivetran sync → hash job → dbt → quality gates → Power BI refresh. Sensors and retries.	Operational overhead. Treat DAGs as product: SLAs, alerts, no hidden cron on the side.
Power BI	How the business already consumes. Import/DirectQuery on Gold only. Analyze in Excel on the certified model.	Logic duplication if measures redefine net revenue or last-click. Dashboards are views, not the model.

GUARDRAILS SO THE STACK DOES NOT FIGHT ITSELF

Each tool has a lane. Crossing lanes is how trust dies.

Fivetran

DOES

Extract + load, schema drift handling, incremental cursors.

DOES NOT

Business rules, UTM meaning, attribution, net revenue.

Databricks

DOES

Hash, identity graph, ugly JSON, heavy Python, feature tables.

DOES NOT

Certified finance/marketing marts that analysts query.

dbt

DOES

Silver/Gold SQL, tests, docs, metric names, SCD2, incremental models.

DOES NOT

Orchestrate Fivetran or replace Airflow SLAs.

Airflow

DOES

Order of operations, retries, sensors, freshness SLAs, paging.

DOES NOT

Contain SELECT logic that belongs in dbt.

Power BI

DOES

Visuals, RLS, certified dataset, Excel live connection.

DOES NOT

Join raw Fivetran tables or invent attribution windows.

PII / hash

DOES

Peppered hash, mapping table, RTBF, secure views.

DOES NOT

Live in Gold facts or get exported to Excel.

ELT for meaning. A thin 'E-hash-L' for privacy.

What I would not do · ETL-heavy

- Clean UTMs inside Fivetran or ADF and drop the raw payload.
- Apply last-click in the ingestion job — you cannot replay a new model.
- Join orders to ad spend in Power BI on raw tables.
- Hash in a BI calculated column (or not at all).
- Let each dashboard pick its own attribution window.

What I would do · ELT + hash

- Bronze = exact source extract (plus Fivetran metadata).
- Hash email/phone in a locked Databricks job before wide access.
- Silver = parse, alias map, identity, SCD2, quality flags.
- Gold = facts/dims + attribution models as data, not DAX.
- Power BI = thin measures (sum, distinct count) on Gold.

Decision rule I would defend in the room

If a rule changes how money or credit is calculated, it belongs in dbt with a test and an owner.
If a rule is visual (colour, chart grain), it belongs in Power BI.
If a rule is 'can we store this identifier at all?', it belongs in the hash / consent path — not in a mart PR.

Stabilise the pipeline at the edge — then map what you cannot block

Ingestion design

- Fivetran for Meta, Google Ads, email, commerce, GA4 BigQuery export if present.
- Incremental / CDC. No nightly full dumps as the default.
- Landing: RAW schema, Fivetran naming, _fivetran_synced, _fivetran_deleted.
- Pixel / webhooks / messy files: Databricks Auto Loader or a small API job.
- Airflow sensor: do not run dbt until sync freshness clears the SLA.
- Backfills are first-class DAGs, not a laptop notebook.

Campaign tagging contract

- Written standard: utm_source / medium / campaign / content / term.
- Allowed values + aliases table (fb → meta, ig → meta, paid-social → paid_social).
- CI-ish check: weekly invalid-UTM report to marketing, not only engineering.
- Paid landing URLs generated from a tool, not typed in Ads Manager.
- Missing UTM → channel = 'unspecified', never silently 'direct'.
- Internal traffic and staff orders flagged, not deleted.

Silver normalisation (dbt) — example grain of the contract

utm_source_raw → utm_source_norm · channel_group (paid_social / paid_search / email / organic / direct / unspecified)
campaign_key = hash(source_norm, campaign_norm, account_id) · dim_campaign SCD2 for name changes
is_valid_utm · is_internal · consent_analytics · event_ts_utc (one timezone, always)
Semi-structured: VARIANT/JSON parsed to typed columns; unknown keys kept in a payload map for replay.

Commercial grain is the order. Credit is a table, not a vibe.

dim_date

Standard date role-playing: order_date, click_date, refund_date.

dim_channel

Paid social, paid search, email, organic, direct, unspecified, internal.

dim_campaign

SCD2. Natural key from platform IDs + normalised UTM campaign.

dim_customer

Hashed customer_sk only. No email. Consent flags as attributes.

fct_touchpoint

One row per impression/click/email-event. Timestamp, campaign_sk, customer_sk, session_id.

fct_order

One row per order. Gross, discount, tax, shipping, net, status, refunded_at.

fct_attribution

order_id × model_name. channel_sk, campaign_sk, attributed_revenue, attributed_weight.

fct_vendor_claimed

What Meta/Google/Email reported. For reconciliation only — never mixed into the KPI.

Keep the hash. Gold never sees raw email.

How we hash

SHA-256(normalised_identifier + pepper). Pepper in Azure Key Vault, not in git. Normalise email (trim, lower) before hash so 'A@B' and 'a@b' collide correctly. Phone in E.164.

PII store

Restricted schema: map_hash (raw → hash), access DE + DPO only. Unity Catalog / Snowflake masks. Secure views for analytics. No SELECT * grants on Bronze PII columns.

RTBF & consent

Right to be forgotten: delete/suppress mapping + tombstone hash in facts (keep financial history aggregated). Consent from CMP joined before activation or detailed profiling. Special-category risk: skin/quiz data stays out of marketing marts.

Clean rooms

JD project. Share hashed or aggregated keys with brand partners — not emails. Snowflake clean rooms or Databricks Clean Rooms. Purpose limitation in the share contract. Audit every query.

Trust is a pipeline with gates — not a Slack thread

Contract tests

dbt

unique + not_null on order_id, touchpoint_id
relationships to dims
accepted_values on channel_group, model_name
conservation: attr revenue = net revenue per model

Freshness & volume

Airflow + dbt source freshness

Fivetran synced_at vs SLA (e.g. Meta < 2h)
Row-count z-score vs 14-day baseline
Zero-row is a fail, not a green empty table

Semantic QA

Recon mart

Vendor claimed vs canonical (tolerance band)
Unspecified UTM %
Refund lag: attributed revenue restated
Direct-share of last-click (sanity)

Observability

Alerts

Pager on SLA miss and conservation break
Dashboard: pipeline, DBU/credit, queue time
dbt exposures so a failed model emails the dashboard owner

TBs/day is an incremental design problem, not a bigger cluster

Partition / cluster

Event and order tables clustered on event_date (and channel where it pays). Never let Power BI scan Bronze.

Incremental everything

Fivetran cursors. dbt incremental + unique_key. Databricks Auto Loader checkpoints. No full rebuild of touchpoints.

Separate compute

Transform warehouse/job cluster ≠ BI warehouse. Finance refresh cannot queue behind a 4TB backfill.

Semi-structured discipline

Parse once in Silver. Gold is typed. Keep raw VARIANT only in Bronze for replay.

Photon / result cache

Databricks Photon or Snowflake result cache + clustering for the certified mart. Pre-aggregate daily ROAS.

Cost guardrails

Auto-suspend. Airflow pool limits. Budget alerts on Fivetran MAR and warehouse credits/DBUs. Prune Fivetran tables we do not model.

Ship trust in slices — do not boil the lake

Days 1–30 · Stabilise

Inventory current dashboards and definitions.
Fivetran → RAW for Meta, Google, email, orders.
UTM alias table + unspecified KPI.
Hash path + PII grants locked.
Stop new Power BI models on raw tables.

Days 31–60 · Canonical model

dbt Silver/Gold: orders, touchpoints, last-click v1.
Conservation tests + freshness SLAs.
Certified Power BI dataset, one default model.
Vendor-claimed recon dashboard.
Airflow DAG replaces hidden refresh jobs.

Days 61–90 · Scale & share

Add linear / position-based models.
Daily spend + ROAS mart.
Identity v1 (deterministic only).
Clean-room pattern for one partner.
Runbook, dbt docs, on-call.

How I would know it worked

- 'Why doesn't this match?' tickets down ≥ 70% after the certified dataset is the only marketing source.
- Unspecified UTM share falling; conservation test green for 14 consecutive days.
- Vendor vs canonical variance explained (window/model), not ignored.
- p95 of the attribution dashboard < 5s (Section 2 will push finance below 3s).

CLOSE

What I want you to remember

01

Multi-tool on purpose

Fivetran ingest, Databricks for hard Python and scale, Snowflake to serve and share, dbt for meaning, Airflow for time, Power BI for views.

02

Logic once

Attribution and revenue are warehouse models with tests. BI does not get a vote on the formula.

03

Hash and recon

Pseudonymous Gold, locked map, RTBF. Vendor numbers sit beside ours — we explain the delta.

Next: Section 2 — the same platform, made self-serve for finance.
Trusted net revenue, fast dashboards, and an end to Excel-as-a-warehouse.